

ETHAN THOMAS

Berkeley, California • 408-329-3621 • ethant@berkeley.edu

EDUCATION

GPA: 4.0

University of California, Berkeley — B.A. Physics, minor in Computer Science

RESEARCH EXPERIENCE

Science Internship Program (SIP) UCSC — Research Intern Summer 2024

- Analyzed RR Lyrae stars and quasar contamination using MARS and FITS datasets.
- Investigated Period–Luminosity–Metallicity (PLZ) relations for pulsating variable stars.
- Developed custom Python workflows with astrophysics libraries for classification.
- Contributed to Feng et al. (2025), arXiv:2512.09795

Science Internship Program (UCSC) — Research Intern Summer 2023

- Exfoliated thin layers of MoS₂ to explore defect-induced magnetism.
- Annealed samples at high temperatures to create magnetism inducing defects

Synopsys Science Fair Projects 2021,2024

- MEMS-based Interventional Radiology Automation: Designed a prototype device to optimize motor placement and mass distribution; implemented Arduino control code with kinematics and Kalman filtering for noise reduction.
- Computer Vision for Smart Refrigeration: Built a CV model to identify household items in fridges and predict expiration dates.

PUBLICATIONS & PRESENTATIONS

- Thomas, E., et al. (2025, Jan). "Analysis of RR Lyrae Stars and Quasar Contamination using MARS and FITS datasets." iPoster presented at the 245th Meeting of the American Astronomical Society (AAS 245), National Harbor, MD. ADS: 2025AAS...24547004S.

TECHNICAL SKILLS

- Programming: Python (NumPy, matplotlib, Astropy), Java, Arduino
- Data Science: Data cleaning, analysis, and representations
- Physics Tools: FITS data, condensed matter foundations